

Important: Retain these instructions

These instructions shall be used by trained service personnel only. If the equipment is used in a manner not specified by these instructions, the protection provided by the equipment may be impaired.

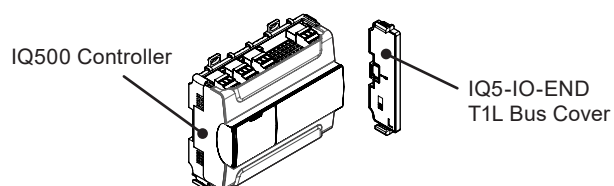
<https://partners.trendcontrols.com>



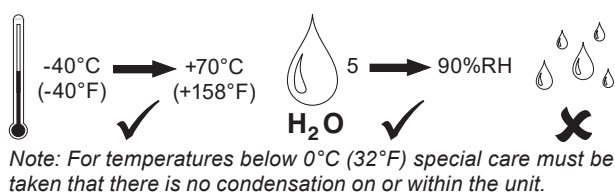
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1 BOX CONTENTS



2 STORING



3 INSTALLATION

Labels used on IQ500

	Caution, consult documentation		Caution, possibility of electric shock
POWER 24V	24 Vac/dc input power connector	RS485 1, 2, 3	RS-485 connector
IQ3/4 IO	IQ4, XCITE I/O bus connector	1 2	Ethernet connectors
	USB local engineering port		RS-485 Terminator switch
	USB (for future use)		Service button
RS485 1 2 3	RS485 port status indicators		
	Trend LAN status indicator	T1L	T1L bus status indicator

It is recommended that the installation should comply with the local electrical safety installation practices (e.g. HSE Memorandum of Guidance on Electricity at Work Regulations 1989, USA National Electric Code).

Any connected devices must be insulated from mains by reinforced insulation.

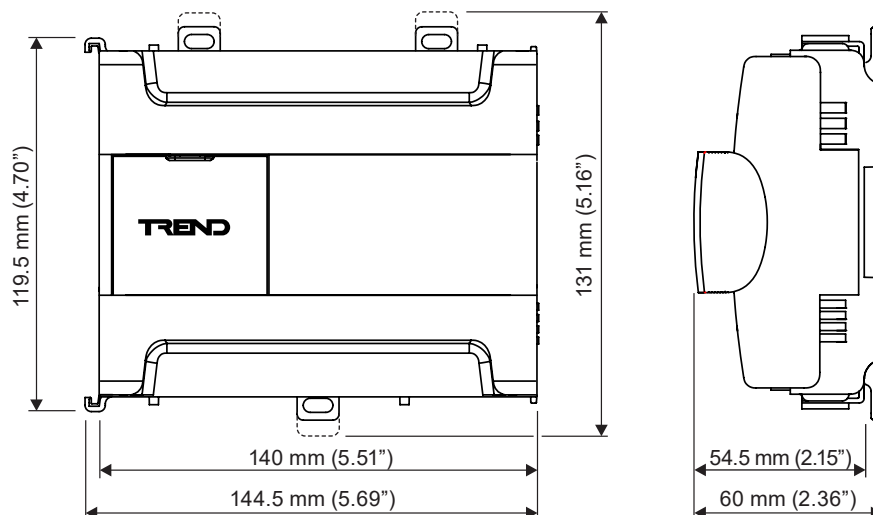
WARNING Removal of cover exposes dangerous voltages.

WARNING To reduce the risk of electrical shock or fire do not interconnect the output of different Class 2 circuits.

	Device status indicator		Service status indicator
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1

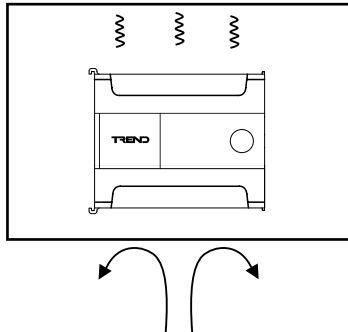
Dimensions



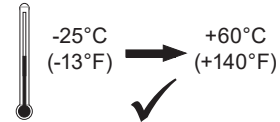
3 INSTALLATION (continued)

2

Mounting Requirements



Protection IP20, NEMA1
 Altitude ≤4000 m (13124 ft)
 Pollution degree 2 (Only non-conducting pollution occurs)



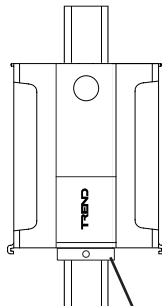
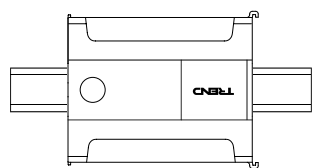
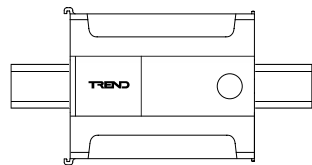
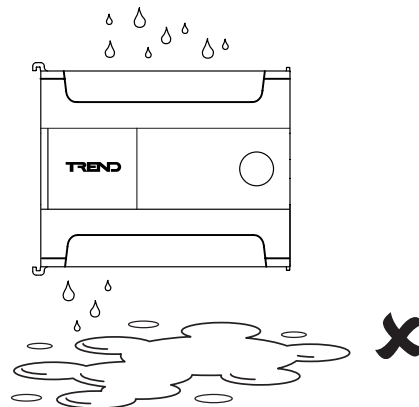
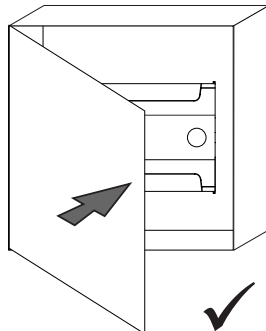
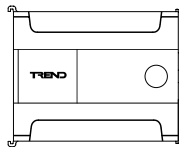
Note: For temperatures below 0°C (32°F) special care must be taken that there is no condensation on or within the unit.



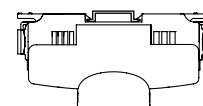
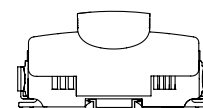
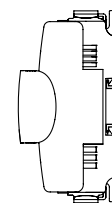
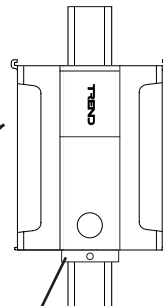
The IQ500 should be installed in an enclosure rated to at least IP20 or equivalent

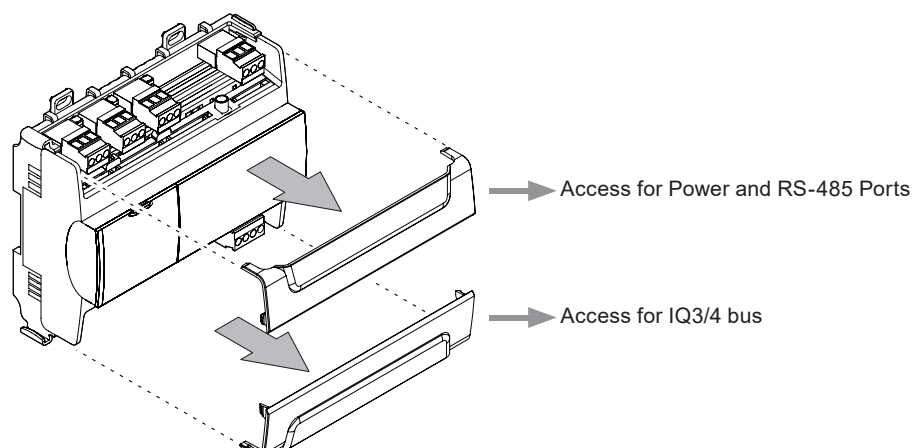
The unit is UL rated as UL60730-1 Automatic electrical controls for household and similar use.

For UL compliance the unit must be mounted in a UL-listed enclosure offering adequate space to maintain the segregation of line voltage field wiring and Class 2 field wiring.

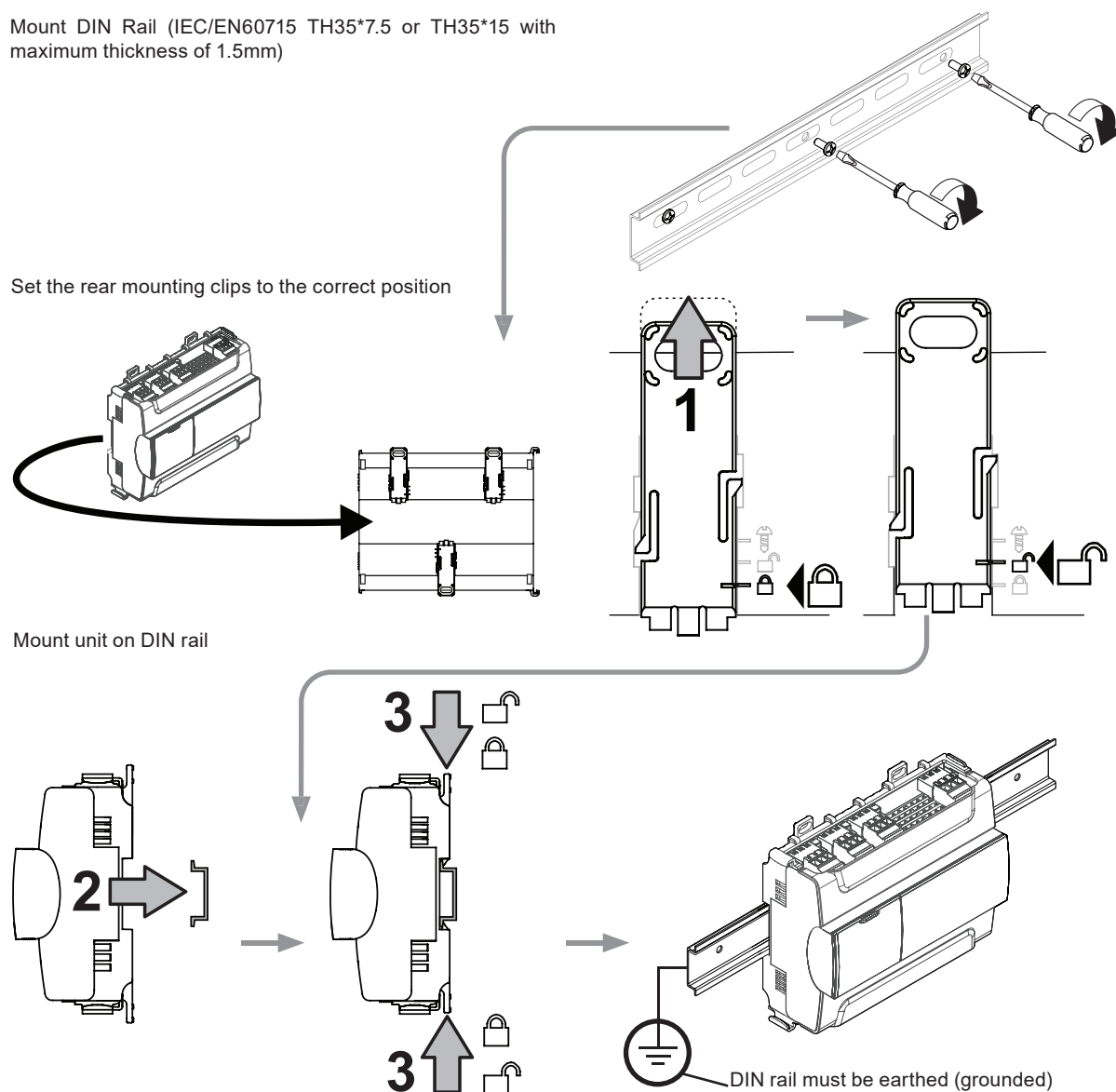


DIN Rail end stop



3 INSTALLATION (continued)**3****Unclip the Terminal Covers****4****Mount Unit**

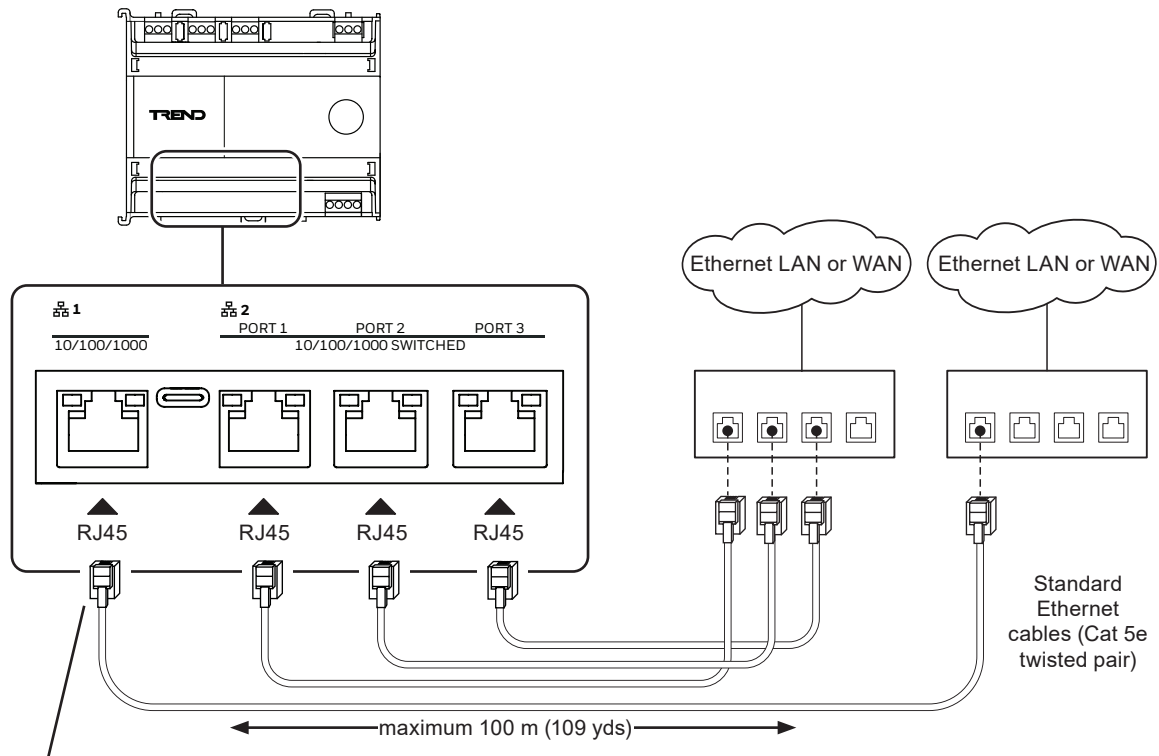
Mount DIN Rail (IEC/EN60715 TH35*7.5 or TH35*15 with maximum thickness of 1.5mm)



3 INSTALLATION (continued)

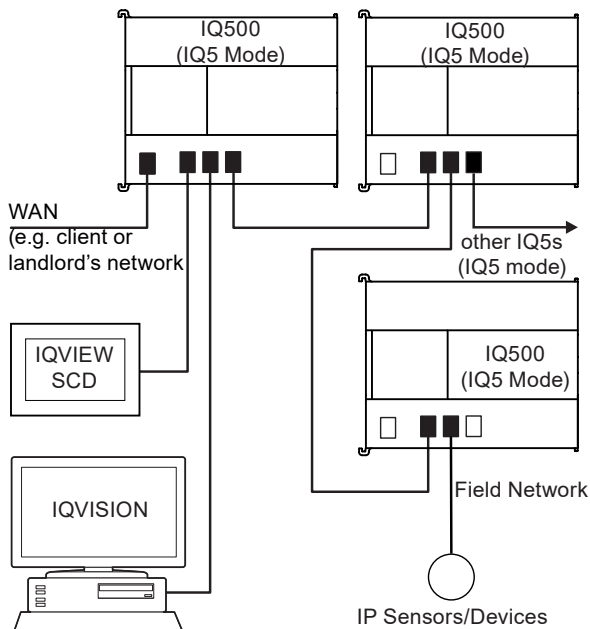
5

Connect Ethernet Network(s)

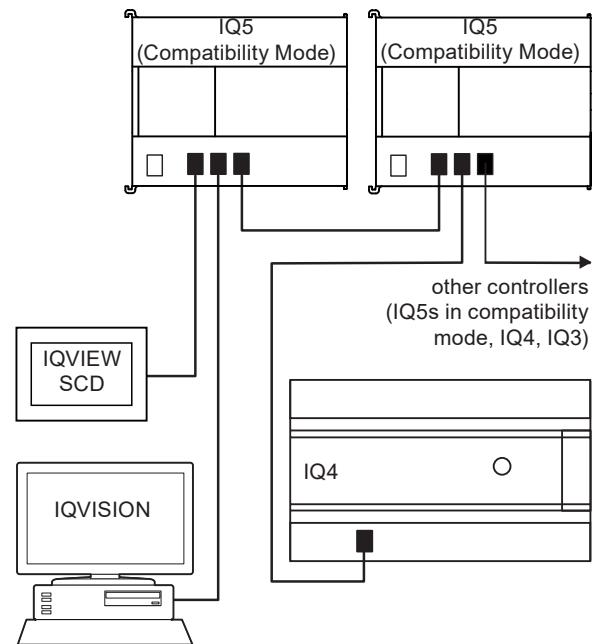


Note: 1 is not available if the IQ500 configured to operate in compatibility mode.

Network Example (IQ5 Mode)



Network Example (Compatibility Mode)

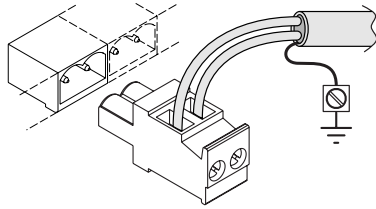


3 INSTALLATION (continued)

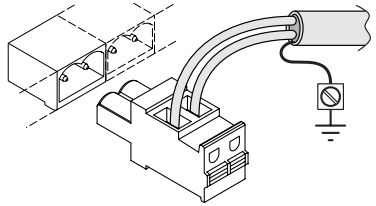
6

Make Connections - Overview

Plug-in connectors with screw terminals (as supplied)



Optional connectors with push-fit terminals



Terminal size: 0.5 to 2.5 mm² (20 to 14 AWG).

Terminal Type	Conductor Size	Stripping Length	Maximum Torque
Screw	IEC: 0.25 to 2.5 mm ² UL: 24 to 12 AWG	6-7 mm	0.5 Nm (4.5 lbf-in)
Push-fit	IEC: 0.2 to 2.5 mm ² UL: 26 to 14 AWG SOL: 24 to 14 AWG STR/SOL	9 mm	n/a

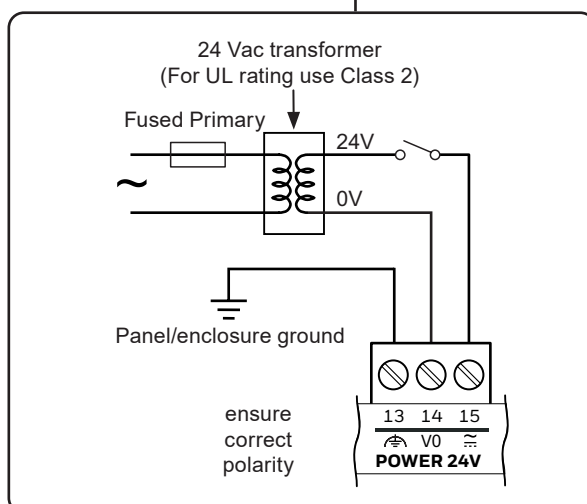
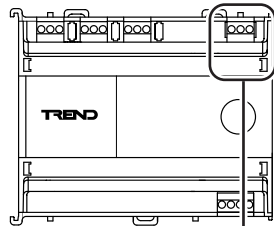
Note: For UL rating use 22 to 14 AWG - Cu only cable.

Note: It is recommended that wire ends are terminated using suitable ferrules, especially when using stranded wire with push-fit terminals.

Connection Type	Go to step...
Power	7
RS-485 Ports	8
T1L Bus	9
IQ3/4 Bus	10

7

Connect Power



This equipment must be earthed (grounded).

Supply rating: 24 Vac $\pm 20\%$, 50/60 Hz, 34VA (1.42A);
24 Vdc $\pm 20\%$, 12.5W (0.52A).

The IQ5 controller can supply 24 Vac/dc to adjacent or remote T1L devices via the power line of the T1L bus. Check that the available current from the controller supply is sufficient to drive the IQ500 and any additional T1L devices.

If more power is required than can be provided by the controller, one or more additional external power supply units must be installed on the T1L bus - see step "9" on page 8.

Refer to the IQ5-IO Modules Data Sheet (TA201481) or IQ5-LAN-ADPT Current Loop LAN Adapter Data Sheet (TA201510) for details of the T1L device power consumption.

Power supply cable must have maximum operating temperature of 80°C or greater.

For UL rating the input power connections must be made using 18 AWG or larger wire rated at least 90°C (194°F).

The primary supply must include a fuse or circuit breaker appropriate for the transformer.

The 24 V supply must include a suitably rated switch in close proximity and be clearly marked as the disconnecting device for the unit.

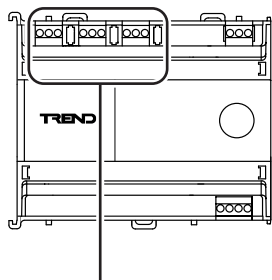
Do not position the equipment so that the disconnecting device is difficult to operate.



3 INSTALLATION (continued)

8

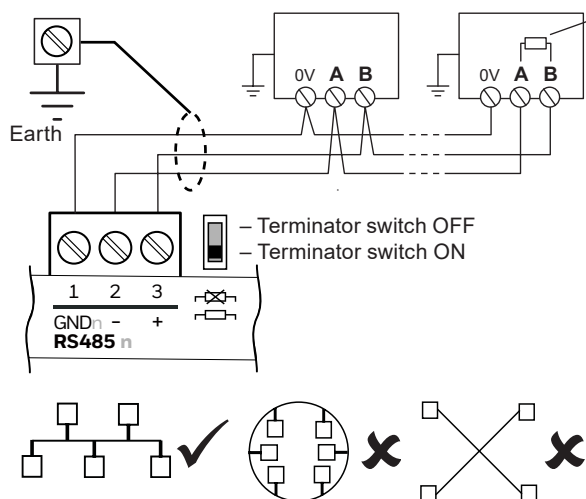
Connect RS-485 Networks (if required)



For use with Modbus, M-Bus, and/or XNC functionality. One port can also be used for an MS/TP LAN of IQECOs.

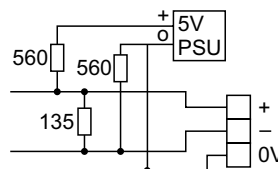
Note: An appropriate converter will be required to enable connection to an RS-232 M-Bus network (e.g. using a PW60) or connection to an RS-485 4-wire network.

Where Modbus, M-Bus or XNC functionality is required this must be enabled by installing an INT licence. Where an MS/TP LAN is required an NC licence must be installed and the controller operated in NC mode (see IQ5, IQ5-IO Installation Instructions - Configuration (TG201483)).



Termination: The bus must be terminated at each end with a resistor matched to the cable characteristic impedance (i.e. $\pm 1\%$, $\frac{1}{4}$ Watt, range 100 to 130 Ω). If the IQ5 is at one end of a 120 Ω cable, switch ON its built-in terminator, otherwise switch it OFF. Termination must be applied at the far end(s) of the cable.

Bias: The IQ500 does not need the bus to be biased as it has a fault tolerant transceiver and applies a small bias. Other devices on the bus may require biasing, in which case it should be provided at one end of the bus using a 5V dc power supply, 2x 560R and 1x 135R, to terminate and bias the bus as shown:



Note: This includes the termination. If connecting to IQ500 do not use the IQ5's built-in terminator.

The GND/0V terminal **MUST** be connected on all devices.

It is not recommended to use the screen as the 0V connection.

Cable screening is recommended for reliable operation in electrically noisy environments. The cable screen should be connected to earth at one point.

Protocol	Device Count	Cable Type	Max Length* metres (feet)
Modbus	32 unit loads	120 Ω twisted pair	900 (2953)
M-Bus	60	120 Ω twisted pair	1000 (3280)
MS/TP	32 unit loads 64 IQECOs**	120 Ω twisted pair	1200 (4000)
XNC	32	depends on chosen application	

* dependent on cable specification and baud rate.

** There may be the IQ5 with up to 64 IQECOs or other manufacturer's devices on the MS/TP trunk. A separate limitation is that the MS/TP segment supports up to 32 'unit' loads. The IQ5 presents a $\frac{1}{4}$ BACnet 'unit' load (ref. EIA-485), as does the IQECO; other manufacturer's devices may have different unit loads.

3 INSTALLATION (continued)

8

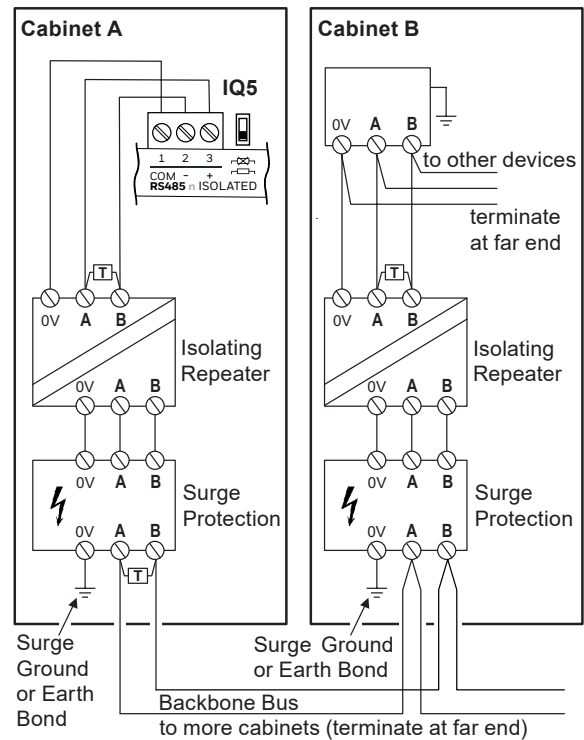
Connect RS-485 Networks (if required)

Earthing (Grounding) & Isolation Requirements

If the IQ500 and other units on the bus are in the same cabinet using the same power supply, each device must have a good physical earth (ground) connection.

If the IQ500 and other units on the bus are in different cabinets or use different power supplies (e.g. different UPS units), the cabinets must be isolated from each other.

If the bus is likely to suffer from surge and grounding problems, surge protection should be added. The isolator should be connected to the earth (ground) of the nearest device, the 0V of the isolator and the surge protector should be connected together, and earth (ground) of the surge protector's exposed side (e.g. backbone bus) should be connected as directly as possible to the surge ground or earth bond. Ensure that terminators are fitted where indicated.



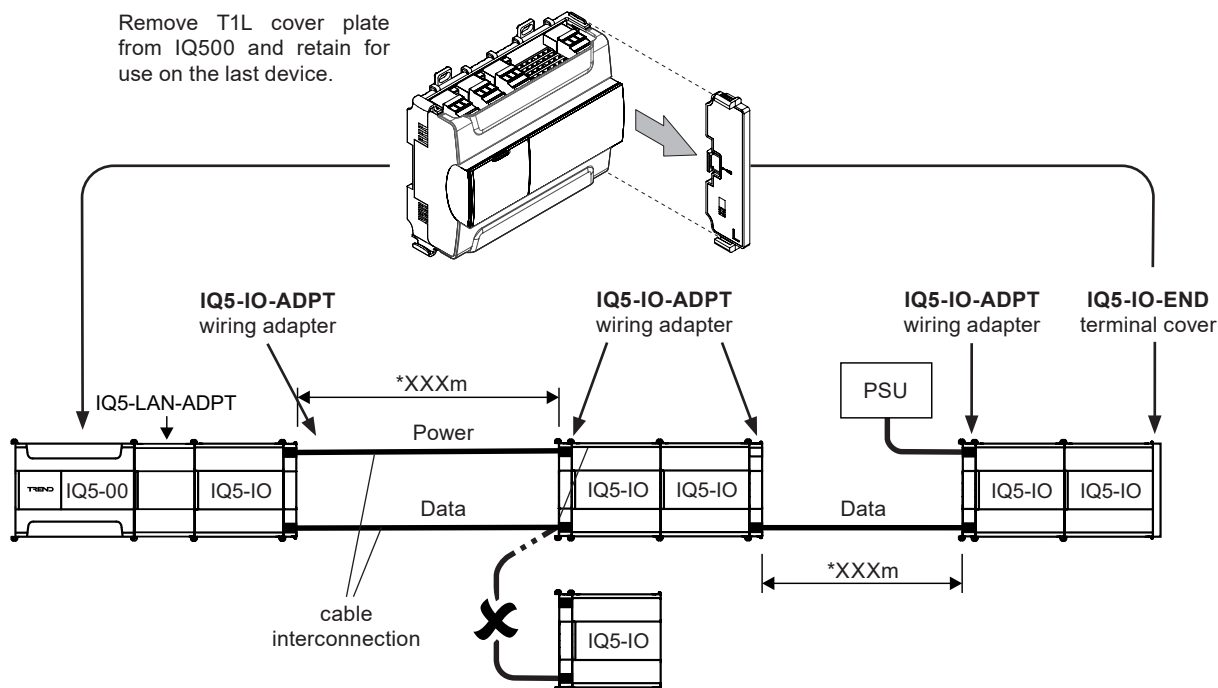
3 INSTALLATION (continued)

9

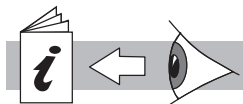
Install and Connect T1L Devices to the T1L Bus (if required)

T1L devices (IQ5-IO modules and the Current Loop LAN Adapter (IQ5-LAN-ADPT)) can be connected to the T1L bus. Depending on the controller licence, up to 300 I/O channels are supported. **Only 1 IQ5-LAN-ADPT can be connected to the controller's T1L bus.**

Remove T1L cover plate from IQ500 and retain for use on the last device.



*XXXm = The recommended $\frac{1}{3}$ of the distance (10 devices off) shown in the table on the next page.



IQ5-IO Installation Instructions - Mounting (TG201484).

IQ5-LAN-ADPT Current Loop LAN Adapter Installation Instructions - Mounting (TG201511).

The IQ5 controller can supply 24 Vac/dc to adjacent or remote T1L devices via the power line of the T1L bus. Check that the available current from the controller supply is sufficient to drive the IQ500 and any additional T1L devices.

If more power is required than can be provided by the controller, one or more additional external power supply units must be installed on the T1L bus.

Note: UL regulations do not allow power from different supplies.

For details of installing an additional external power supply units must be installed on the T1L bus. see the mounting instructions for the T1L device.

Note: It may also be necessary to split the T1L bus supply to avoid exceeding the maximum 2 A flowing through the IQ5 or T1L devices.

When designing a T1L network which goes beyond the MCC panel for I/O modules, the distance between T1L devices is impacted by several factors:

- Cable characteristics – Each cable with it's specific characteristics of impedance, capacitance, insertion loss and return loss will have an impact on transmission distance (see table on next page).
- Quality of cable terminations and installation
- Number of devices on the T1L network but switched off. T1L is a point-to-point network, each device receives the messages, reviews and then transmits them on to it's neighbour. A T1L device has network by-pass relays which when powered down allow the communications to pass through the offline device and therefore ensure network communications are maintained. These offline devices reduce the maximum distance (see table on next page).
- The T1L network standard is defined to provide up to 1,000 meters.
- Bus voltage must not be allowed to drop below 19V at full load.

Note: When designing the network, you must be aware of the power load of each T1L device as this may require you to break the network to add additional power supplies.

3 INSTALLATION (continued)

9

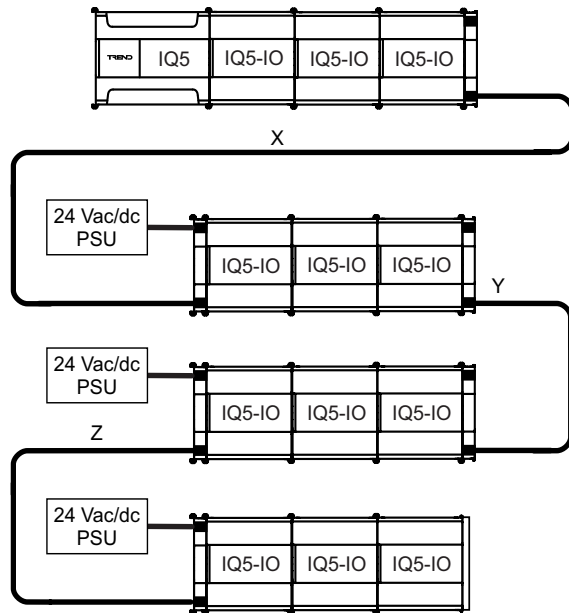
Install and Connect T1L Devices to the T1L Bus (if required) (continued)

Trend have tested the T1L devices and identified the following design rules should be considered:

- A maximum of 10 offline devices between active devices and the maximum distance between active devices, is as defined by lab cable testing or theoretical calculations based on cable characteristics.

Note: If you have remote devices ensure separate power sources are used to avoid single points of failure.

When designing a system with remote IO you need to be aware that separately powered panels with T1L devices may be switched off and hence the distance between T1L devices may now be $X + Y$ or even $X + Y + Z$. For this reason, we recommend the distance between panels of IO be no more than the recommended $\frac{1}{3}$ of the distance (10 devices off) shown in the table.



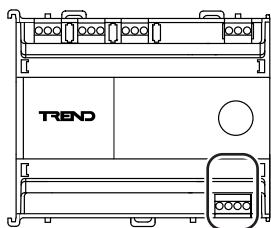
The table below identifies cables that have been tested and the distances which can be achieved between active devices assuming up to 10 offline devices are between them.

Cable	Typical Uses	Cable Characteristics	Max Distance (between working nodes)	Max Distance (10 devices off)	Recommended $\frac{1}{3}$ Distance (10 devices off)
74040NH – Belden	T1L- Long distance harsh environments	2 core solid 18AWG, SF/UTP Shielded and foil, unshielded twisted pair.	1000m	900m	300m
8471 – Belden	LON	2 core stranded/tinned 16AWG, unshielded cable.	560m	504m	168m
9841 – Belden TP/1/1/24/200/HF-600V	MSTP (Serial)	2 core stranded/tinned 24AWG, Foil shield & drain wire, twisted pair.	400m	360m	200m
8723NH – Belden TP/2/2/22/200/HF-600V	Trend 4 wire LAN	4 core, 2 pairs, stranded/tinned, 22AWG, Foil shield & drain wire, twisted pairs.	160m	144m	48m
8761NH – Belden TP/1/1/22/200/HF-600V	Trend 2 wire LAN	2 core stranded/tinned, 22 AWG, Foil shield & drain wire, twisted pairs.	320m	288m	96m
5501UE 0081000 – Belden/BAV	Security, speaker, PA, & telephone systems	3 core stranded bare copper, 22 AWG, no shield or twist.	600m	540m	180m
82836 – Helukabel	Profibus - Industrial Ethernet	2 core solid, 18AWG Foil + braided screen twisted pair.	800m	720m	240m
3076F – Belden	Harsh environment digital and serial two way communication	2 core solid 18AWG, Shielded and foil, unshielded twisted pairs.	428m	385m	128m
Helukabel J-Y(ST)YLG	Telecommunications & Fire Alarm Cable (Fire Warning Cable)	Multicore solid bare copper, 20AWG (0.8mm), foil wrapped.	320m	288m	96m
Cat5/Cat6	Standard wired IT network cable	8 core, 4 pairs, solid bare copper, 23AWG, twisted pairs.	720m	648m	217m

3 INSTALLATION (continued)

10

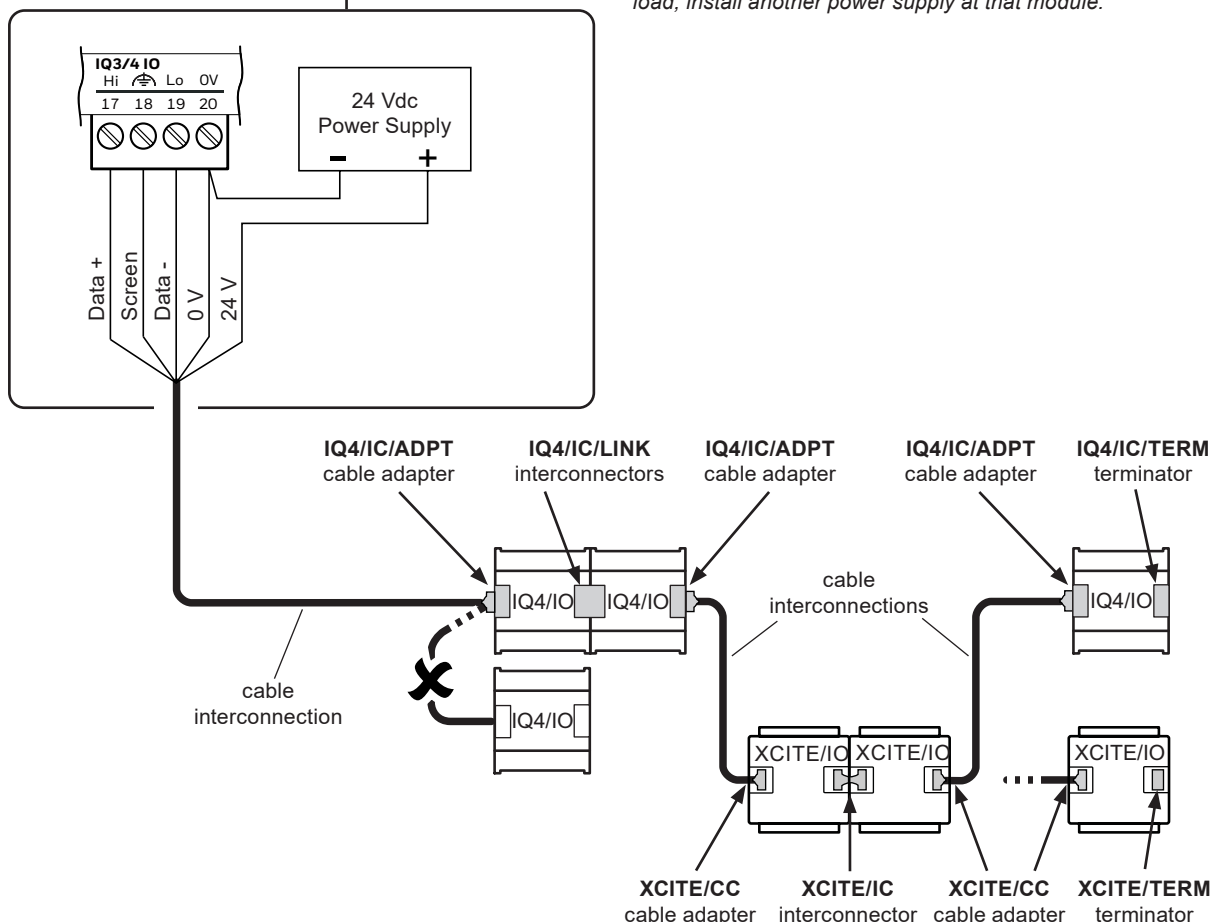
Install and Connect IQ4/IO or XCITE/IO Modules to the IQ3/4 Bus (if required)



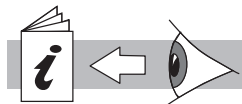
The IQ3/4 bus provides compatibility with I/O modules from the IQ4/IO and XCITE/IO ranges. Up to 192 I/O channels are supported on this bus.

Note: To use this bus it must be enabled by installing a CAN licence. See IQ500 Data Sheet (TA201480) for further details.

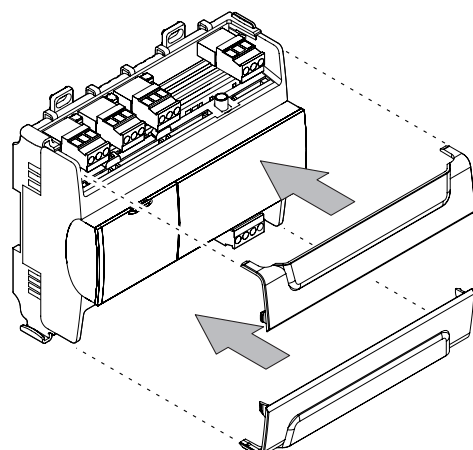
Note: No power is available via this connector. A separate 24 Vdc power supply must be provided. For all installations, if the bus voltage at an I/O module drops below 19.2 V at full load, install another power supply at that module.



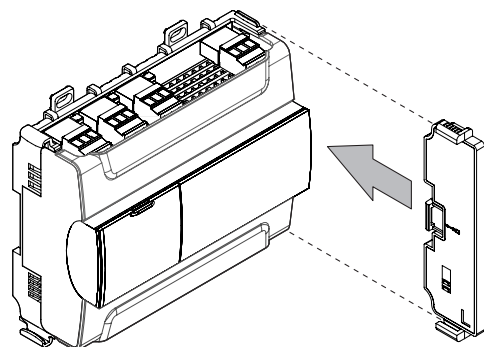
Controller and IO Module Configuration	Maximum I/O bus length depending on cable type (excluding rigid interconnectors)		Maximum No. of Modules
	Belden 3084A	Belden 7895A	
IQ500 controller with IQ4/IO modules only	Total bus length up to 100 m (109 yards) or Total bus length up to 300 m (328 yards) if IQ4/IO modules are within 100m (109 yards) of a power supply	total bus length up to 300 m (328 yards)	30
IQ500 controller with IQ4/IO modules and/or XCITE/IO modules	total bus length up to 30 m (33 yards) <i>Note: Maximum 10 m (11 yards) for certain system configurations - for details check the XCITE/IO Modules Data Sheet (TA201352) and XCITE/IO Installation Instructions - Mounting (TG200627).</i>		15



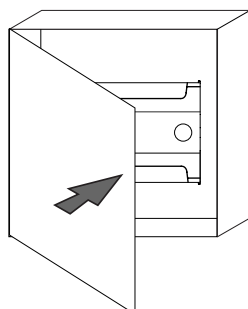
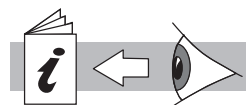
IQ4/IO Installation Instructions - Mounting (TG201342)
XCITE/IO Installation Instructions - Mounting (TG200627)

3 INSTALLATION (continued)**11****Refit Terminal Covers**

Note: Spare covers are available to order (e.g. IQ5-TCVR-140-10)

12**Fit IO Terminal Cover**

Note: Spare covers are available to order (e.g. IQ5-IO-END-10)

13**Close Panel / Enclosure****14****Configure IQ5 Controller and I/O Modules**

IQ500, IQ500-IO Installation Instructions - Configuring (TG201483)
IQ4/IO/.. Installation Instructions - Configuring (TG201343)
XCITE/IO/.. Installation Instructions - Configuring (TG201161)

4 FIELD MAINTENANCE

The IQ500 require no routine maintenance.

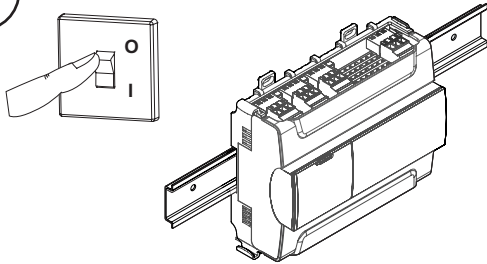


WARNING: Contains no serviceable parts. Do not attempt to open the unit. Failure to comply may cause damage to the unit.

5 REMOVING CONTROLLER FROM DIN RAIL

1

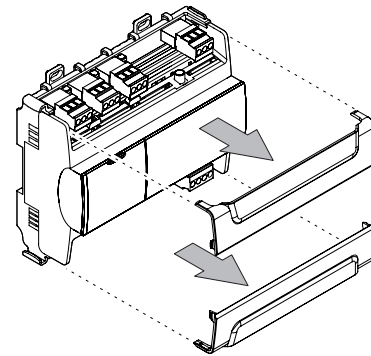
Isolate Power (including IO supplies)



Warning: Hazardous voltages may be present on I/O terminals. Isolate before touching.

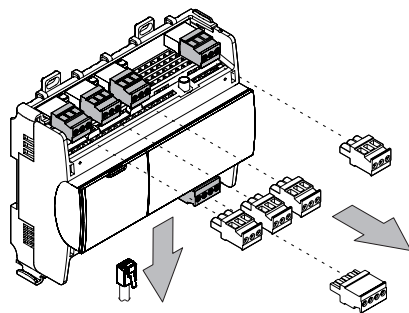
2

Remove Terminal Covers



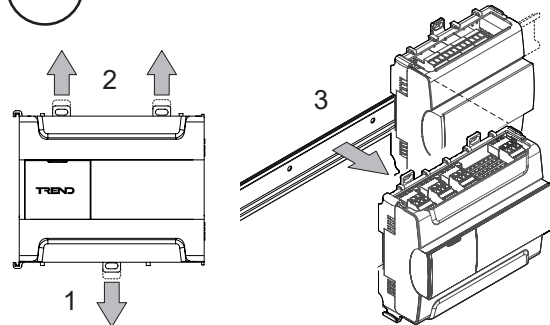
3

Unplug Wired Connectors



4

Unclip from DIN Rail



6 DISPOSAL



WEEE Directive:

At the end of their useful life the packaging and product should be disposed of by a suitable recycling centre.

Do not dispose of with normal household waste.
Do not burn.

7 CHINA HAZARDOUS SUBSTANCES TABLE

产品中有害物质的名称及含有的信息表										
部件名称 Component Name	List of Hazardous Substances in the Product and Relevant Information									
	铅 (Pb) Lead (Pb)	汞 (Hg) Mercury (Hg)	镉 (Cd) Cadmium (Cd)	六价铬 (Cr6+) Chromium VI Compounds (Cr6+)	多溴联苯 (PBB) Polybrominated Biphenyls (PBB)	多溴二苯醚 (PBDE) Polybrominated Diphenyl Ethers (PBDE)	邻苯二甲酸二 (2-乙基)己 酯 (DEHP) Bis(2- ethylhexyl) phthalate (DEHP)	邻苯二甲 酸丁苯酯 (BBP) Butyl benzyl phthalate (BBP)	邻苯二甲酸 二正丁酯 (DBP) Dibutyl phthalate (DBP)	邻苯二甲酸二 异丁酯 (DIBP) Diisobutyl phthalate (DIBP)
PCBA	X	O	O	O	O	O	O	O	O	O
本表格依据 SJ/T 11364 的规定编制。 This table is prepared in accordance with the provisions of SJ/T 11364.										
O 表示该有害物质在该部件所有均质材料中的含量均在 GB/T 26572 规定的限量要求以下。 Indicates that said hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement of GBT 26572.										
X 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T 26572 规定的限量要求。 Indicates that said hazardous substance contained in all of the homogeneous materials for this part is above the limit requirement of GBT 26572.										
环保使用 年限 EPUP	10									
未列入表内的其他部件, 皆不含任何超出限量要求的限制使用物质。 All other components, not listed in the table, do not contain restricted substances above the threshold level										

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